Passivity of a Dynamic Environment in Telecontrol Systems

Ho Duc Tho and Takanori Miyoshi
Nagaoka University of Technology, 1603-1 Kamitomioka Nagaoka, Niigata, Japan 940-2188
Emails: hoductho@vos.nagaokaut.ac.jp, miyoshi@vos.nagaokaut.ac.jp.

***Abstract*— In this brief note, we will show that the dynamic environment presenting in telecontrol systems is passive regardless of the number of contacts and contacts' stiffness if it is modeled as a network of connected spring-mass elements.**

I. Introduction

Passivity is an important concept in the modern control theory. In a passive system, the energy increasing within the system is bounded by the supplied energy from outside [1]. The passivity belongs to the school of thought which emphasizes on analyzing the input-output relation of the system. Under appropriate conditions, passivity implies $\mathcal{L}_2$ and Lyapunov stability. Passivity has two merits. First, a passive system is easy to control, namely that it can be stabilized by a simple PD control law plus the zero-state observability condition. Second, the passive theorem states that the negative feedback connection of two passive systems is passive, which is a result of viewing the whole system as a process of energy exchange. Indeed, this is the most attractive feature of the control system design based on passivity. The reason is that it allows the control design of a complex system to be decomposed into the control designs for simpler subsystems connected via a network.

The above-mentioned invariant property of passivity under a feedback connection is particularly useful in the field of telerobotics, where the time-delay resulting from a distant communication poses a critical problem to the stability of force-reflecting teleoperations [2]–[4]. A common telecontrol system is a network consisting of five elements: human operator, master robot, communication channel, slave robot, and environment. Using the passivity formalism, Anderson and Spong [2] designed control laws to passify both master and slave robots, in which two robots track the velocity of one another (transparency). Moreover, the master robot tries to imitate the environment force of the slave side to provide the force information to the human operator (telepresence). A passive communication law was then designed to circumvent the time-delay issue, where the passivity was proved by scattering theory. Moreover, it has been pointed out that it is reasonable to consider the human operator as a passive system [5]. The passivity of the last element, i.e., the environment, however, needs to be checked/ensured. The environment is often physically real in which objects interacting with slave robots could be static (e.g., wall, ground) or dynamic (e.g., human, nearby robots). Nonetheless, in some cases, a virtual environment called motion server [6] is created to allow arbitrary many distant clients to share the feeling of forces over the Internet when they concurrently interact with a dynamic virtual environment, and thus, multilateral control is termed.

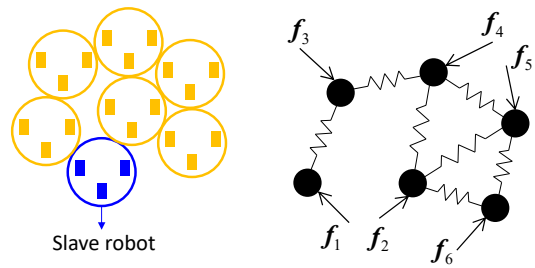

Fig. 1. A swarm of green robots (left) is considered as a dynamic environment, which is modeled by a connected spring-mass system (right).

Contacts within the environment are often modeled by spring-mass-damper elements [7], [8]. Here, we will consider the worst-case scenario where the environment has no internal damping. Under such a consideration, the environment is intuitively passive, yet a rigorous proof for such a claim needs to be sought. In particular, multiple contacts might exist in dynamic environments (see Fig. 1) making the passivity proof nontrivial. Furthermore, during the operation, dynamic elements of the environment frequently make or break contacts, so the spring constants of the contacts vary with time. This adds extra complexity to the problem. Thus, the passive invariance of dynamic environments under the change of spring constants should be proven. In this paper, we will show that if the dynamic environment is assumed to be a globally connected spring-mass system (see Definition 1), the above-mentioned passivity invariance can always be achieved regardless of spring constants and the number of contacts.

II. Main result

For the sake of simplicity, we only consider a planar dynamic environment. The extension to the n-dimensional rigid-body case with gravity effect is straightforward.

Definition 1. A globally connected spring-mass system is a set of N masses, in which the weight of the i-th mass is m_i. Every two masses i-th and j-th of the system are connected by a spring with a spring constant $k_{ij} \geq 0$. Moreover, there is a force $\boldsymbol{f}_i \in \mathbb{R}^2$ acting on the mass i-th, $\forall i = 1, \ldots, N$.

An example of the globally connected spring-mass system with $N = 6$ is shown in Fig. 2.

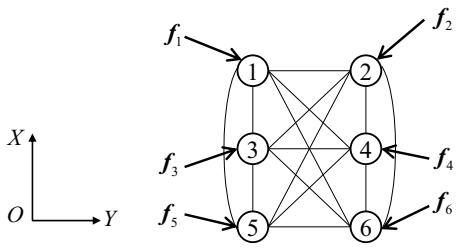

Fig. 2. A globally connected spring-mass system with $N = 6$.

Assumption 1. The dynamic environment with which the slave robot interacts can be modeled as a globally connected spring-mass system defined in Definition 1.

To prove that a dynamic environment satisfying Assumption 1 is passive, we need the following proposition.

Proposition 1. Let $\boldsymbol{S}$ be a holonomic mechanical system whose (independent) generalized coordinate and associating generalized force are $\boldsymbol{q} = [q_1\ q_2\ \dots\ q_n]^\top \in \mathbb{R}^n$ and $\boldsymbol{F} \in \mathbb{R}^n$ respectively. Assuming that the potential energy U of the system is only dependent on $\boldsymbol{q}$ and the reference point is chosen such that U is non-negative. Then, the system $\boldsymbol{S}$ is passive if the kinetic energy $T(\boldsymbol{q}, \dot{\boldsymbol{q}})$ of the system satisfies

$$\dot{\boldsymbol{q}}^\top \left[\frac{d}{dt}\left(\frac{\partial T(\boldsymbol{q}, \dot{\boldsymbol{q}})}{\partial \dot{\boldsymbol{q}}} \right) - 2\frac{\partial T(\boldsymbol{q}, \dot{\boldsymbol{q}})}{\partial \boldsymbol{q}} \right] = \ddot{\boldsymbol{q}}^\top \frac{\partial T(\boldsymbol{q}, \dot{\boldsymbol{q}})}{\partial \dot{\boldsymbol{q}}}. \quad (1)$$

Proof. The Lagrangian associating with the system is

$$L = T(\boldsymbol{q}, \dot{\boldsymbol{q}}) - U(\boldsymbol{q}). \quad (2)$$

Since the system is holonomic and the generalized coordinates are independent, the motion of the system satisfies the Euler-Lagrange equation [9]

$$\begin{aligned} &\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\boldsymbol{q}}} \right) - \frac{\partial L}{\partial \boldsymbol{q}} = \boldsymbol{F}. \\ \Rightarrow\ &\frac{\partial U(\boldsymbol{q})}{\partial \boldsymbol{q}} = \boldsymbol{F} - \frac{d}{dt}\left(\frac{\partial T(\boldsymbol{q}, \dot{\boldsymbol{q}})}{\partial \dot{\boldsymbol{q}}} \right) + \frac{\partial T(\boldsymbol{q}, \dot{\boldsymbol{q}})}{\partial \boldsymbol{q}}. \end{aligned} \quad (3)$$

To show the passivity of the system, the following semipositive definite storage function is chosen

$$V = T(\boldsymbol{q}, \dot{\boldsymbol{q}}) + U(\boldsymbol{q}). \quad (4)$$

The derivative of V can be computed by

$$\dot{V} = \dot{T}(\boldsymbol{q}, \dot{\boldsymbol{q}}) + \dot{U}(\boldsymbol{q}) = \dot{\boldsymbol{q}}^\top \frac{\partial T}{\partial \boldsymbol{q}} + \ddot{\boldsymbol{q}}^\top \frac{\partial T}{\partial \dot{\boldsymbol{q}}} + \dot{\boldsymbol{q}}^\top \frac{\partial U}{\partial \boldsymbol{q}}. \quad (5)$$

Substituting (3) into (5) yields

$$\dot{V} = \dot{\boldsymbol{q}}^\top \boldsymbol{F} + 2\dot{\boldsymbol{q}}^\top \frac{\partial T}{\partial \boldsymbol{q}} + \ddot{\boldsymbol{q}}^\top \frac{\partial T}{\partial \dot{\boldsymbol{q}}} - \dot{\boldsymbol{q}}^\top \frac{d}{dt}\left(\frac{\partial T}{\partial \dot{\boldsymbol{q}}} \right). \quad (6)$$

If (1) is fulfilled, the following result can be obtained

$$\dot{V} = \dot{\boldsymbol{q}}^\top \boldsymbol{F}, \quad (7)$$

which implies that the system is passive [1, p. 236] with the input-output pair $(\boldsymbol{F}, \dot{\boldsymbol{q}})$. This completes the proof. □

We are now ready to prove the main result of the paper.

Theorem 1. A dynamic environment satisfying Assumption 1 is passive regardless of the spring constants and the number of contacts.

Proof. Denote the position of the mass i-th in the OXY coordinate system as $\boldsymbol{q}_i = [x_i\ y_i]^\top$. The system can be completely described by the following vector of independent Cartesian coordinates

$$\boldsymbol{q} = \left[\boldsymbol{q}_1^\top\ \dots\ \boldsymbol{q}_N^\top\right]^\top = [x_1\ y_1\ \dots\ x_N\ y_N]^\top \in \mathbb{R}^{2N}. \quad (8)$$

Let f_{ix} and f_{iy} be the projections of $\boldsymbol{f}_i$. The generalized force vector associating with the generalized coordinate $\boldsymbol{q}$ is

$$\boldsymbol{F} = \left[\boldsymbol{f}_1^\top\ \dots\ \boldsymbol{f}_N^\top\right]^\top = [f_{1x}\ f_{1y}\ \dots\ f_{Nx}\ f_{Ny}]^\top \in \mathbb{R}^{2N}. \quad (9)$$

The kinetic energy of the system is

$$T = \frac{1}{2}\sum_{i=1}^{N} m_i \left(\dot{x}_i^2 + \dot{y}_i^2\right). \quad (10)$$

The potential energy U of the system is the summation of all potential energy stored in the springs connecting two arbitrary masses. For example, the potential energy of the spring connecting two masses i-th and j-th is given by

$$U_{ij} = \frac{1}{2}k_{ij}||\boldsymbol{q}_i - \boldsymbol{q}_j||_2^2. \quad (11)$$

Therefore, U_{ij} is only dependent on $\boldsymbol{q}$, and so is the total potential energy U. It is easy to verify that the globally connected spring-mass system satisfies all conditions in Proposition 1, and hence, it is passive. □

III. Conclusion

In this paper, the passivity of a dynamic environment in a telecontrol system is studied. It has been shown that if the environment is modeled by a network of globally connected spring-mass elements, the environment is passive regardless of the number of contacts and contacts' stiffness.

References

[1] H. Khalil, *Nonlinear Systems*, 3rd ed. Upper Saddle River, NJ: Pearson, 2002.

[2] R. J. Anderson and M. W. Spong, "Bilateral control of teleoperators with time delay," *IEEE Trans. on Automatic Control*, vol. 34, no. 5, pp. 494–501, 1989.

[3] E. Nuno, L. Basanez, R. Ortega, and M. W. Spong, "Position tracking for non-linear teleoperators with variable time delay," *The Int. J. of Robotics Research*, vol. 28, no. 7, pp. 895–910, 2009.

[4] T. Hatanaka, N. Chopra, M. Fujita, and M. W. Spong, *Passivity-Based Control and Estimation in Networked Robotics*. Switzerland: Springer, 2015.

[5] R. J. Anderson and M. W. Spong, "Asymptotic stability for force reflecting teleoperators with time delay," *The Int. J. of Robotics Research*, vol. 11, no. 2, pp. 135–149, 1992.

[6] T. Miyoshi, T. Imamura, and K. Terashima, "Stability analysis via iqc for multilateral tele-control and application to multi-client/multi-coupling physical model system," in *41st Annu. Conf. of the IEEE Ind. Elec. Soc.*, 2015, pp. 3388–3392.

[7] S. Eppinger and W. Seering, "Introduction to dynamic models for robot force control," *IEEE Control Systems Magazine*, vol. 7, no. 2, pp. 48–52, 1987.

[8] F. Lange, W. Bertleff, and M. Suppa, "Force and trajectory control of industrial robots in stiff contact," in *IEEE Int. Conf. on Robotics and Automation*, 2013, pp. 2927–2934.

[9] D. T. Greenwood, *Advanced Dynamics*. UK: Cambridge University Press, 2003.